

Sebastian Holtedahl Castedo

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EDUCATION

Ecole Normale Supérieure, FR Oct 2023 – Sept 2026
PhD in Statistical Physics under the supervision of Rémi Monasson.

- Investigating the trade-offs between performance and energy consumption in natural and artificial neural networks to uncover principles of computational efficiency. Developing statistical models to analyse high-dimensional neural data, using theoretical tools.
- Published: *Speed Vascular Patterns in the Spatial Navigation System* [Cell Reports](#)
- In prep: *Metabolic Constraints Reshape Neural Noise Geometry for Efficient Coding.*
- Invited talks: PINTS 2025, PBPCD 2025, UK Neural Computation 2025
- Posters presented at conferences: Eresfjord School (Math Methods for Comp Neuro), Cargèse (Maturation & Plasticity in NN) and Int'l Conf. on Mathematics of NeuroAI.

The University of Manchester, UK Sept 2019 – June 2023
MPhys Theoretical Physics with Study in Europe.

- First Class Honours (82%). Third year exchange programme at Sorbonne Université in Paris.

SKILLS

Language: Fluent in English, Spanish, Norwegian and French.
Technical: Python (Pytorch), C++, Fortran. MCMC and Optimisation methods.

WORK EXPERIENCE

The University of Manchester, UK Sept 2022 – June 2023
Master Thesis in Statistical Physics under the supervision of Tobias Galla

- Analysed the impact of structured correlations on the dynamics of disordered systems using mean-field theory, with applications to theoretical ecology. Results published in *Phys. Rev. E* [arXiv:2409.12751](#).

Institute for Cross-Disciplinary Physics and Complex Systems, Palma ES Jul 2022 - Aug 2022
Summer Undergraduate Research Fellowship (SURF)

- Applied Variational Autoencoders to microbiota data for childhood obesity prediction.

Laboratoire de Physique Théorique et Hautes Energies, Paris FR Apr 2022 - Jun 2022
Master's Internship under the supervision of Leticia Cugliandolo

- Theoretical analysis of a stochastic system using the Langevin and Fokker-Planck formalisms. Results published in *Phys. Rev. B* [arXiv:2205.10418](#).

Fermi National Accelerator Laboratory, Chicago US Jul 2021
Quantum Computing Internship for Physics Undergraduates

- Implemented quantum algorithms (QFT, Grover's, QPE) using Python & Qiskit and covered theoretical foundations of quantum information theory.

ADDITIONAL EXPERIENCE & INTERESTS

- Accenture Horizons Internship (Sept 2018 – May 2019):** Developed optimisation tools for a government technology delivery project.
- Social Impact:** Helped found an educational charity mentoring under-represented students; co-founded a student-housing cooperative; co-authored an open-source deforestation-mitigation report for COP26.